

Financial Modeling Challenge

*A Bangladeshi logistics company, Apex Delivery Ltd., wants to purchase a fleet of new electric delivery vans. You need to build a financial model to evaluate the **Loan Payments** for financing this purchase and determine the project's profitability using a **DCF Model**.*

Part 1: The Loan Amortization Problem

To buy the delivery vans, the company takes a commercial vehicle loan from a bank in Dhaka. Prepare the Loan Amortization Schedule.

Loan Amount (Principal): **BDT 1,200,000**

Annual Interest Rate: **10.5%**

Loan Tenure: **4 Years**

Part 2: The DCF Valuation Problem

The fleet of vans is expected to generate revenue over the next 4 years. The company wants to know if this investment is worth it.

Initial Investment (Year 0 Cash Outflow): **BDT 1,200,000** (The cost of the vans)

Expected Annual Cash Inflows (Profits after maintenance):

Year 1: **BDT 400,000**

Year 2: **BDT 500,000**

Year 3: **BDT 450,000**

Year 4: **BDT 350,000**

Discount Rate (Cost of Capital): **11%**